

A Question Types

In Section 3.2, we predefine four question types for the convenience of annotation, with an example of each type provided in Figure 5.

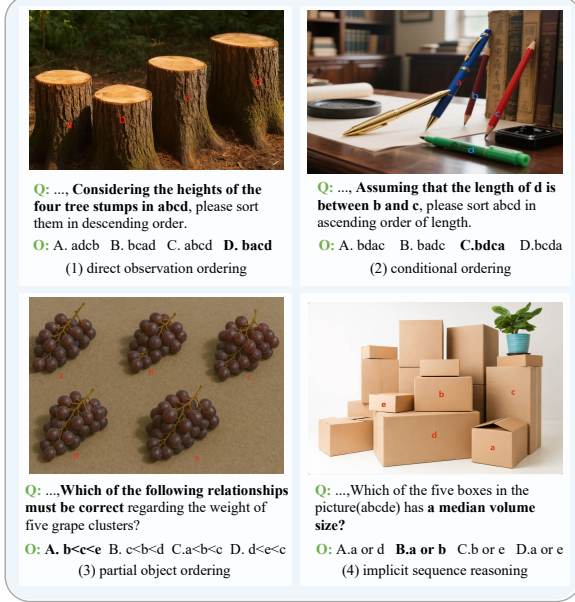


Figure 5: Examples of four question types.

B Implementation Details

The parameter settings of the open-source MLLMs used in Table 3 are provided in Table 7. In addition, the task prompts used for the MLLMs are listed in Table 8.

Model	EP	BS	ES	LR	Opt	LR
Deepseek-VL	6	8	-	2e-5	AdamW	cos
MiniCPM-v4.5	6	8	-	2e-5	AdamW	cos
Llama-3.2	6	8	-	2e-5	AdamW	cos
Llava-ov-2	6	8	-	2e-5	AdamW	cos
Phi-4	6	8	-	2e-5	AdamW	cos
Janus-Pro	2	8	-	5e-6	AdamW	cos
Qwen3.5	6	8	-	2e-5	AdamW	cos
InternVL3.5	6	8	-	2e-5	AdamW	cos
mPLUG-Owl3	6	8	-	2e-5	AdamW	cos

Table 7: Hyper-parameter setting for open-source MLLMs. “EP”, “BS”, “ES”, “LR”, “Opt”, and “LR” denote “Epochs”, “Batch Size”, “Early Stop”, “Learning Rate”, “Optimizer”, and “Learning Rate Scheduler”, respectively.

Task prompt

Open-source MLLMs and Zero-shot for Closed-source MLLMs

You are currently a senior expert in sequence problems, focusing on specific research topics such as time, space, length, quantity, emotion, symmetry, logic, etc. You can sort their attributes, such as length, order, size, quantity, and strength. Given one or more images and a sorting problem, some questions can be answered directly, some questions require inference, and others can only choose relatively reasonable answers. Your task is to answer these questions from a human perspective and output the correct answers without any explanation. Please note that you only need to select one option from all options, and your response should only include the option letter (A, B, C, or D) and not any other text. \n Input: Image: <image> \n Question: {question}, Options: {options}. \n Output: *Note that for InstructBlip-7B, the prompt excludes the image component.

Few-shot for Closed-source MLLMs

You are currently a senior expert in sequence problems, focusing on specific research topics such as time, space, length, quantity, emotion, symmetry, logic, etc. You can sort their attributes, such as length, order, size, quantity, and strength. Given one or more images and a sorting problem, some questions can be answered directly, some questions require inference, and others can only choose relatively reasonable answers. Your task is to answer these questions from a human perspective and output the correct answers without any explanation. Please note that you only need to select one option from all options, and your response should only include the option letter (A, B, C, or D) and not any other text. \n Given the following 3 examples to learn the visual reasoning task: \n Example1: \n Image: <image> \n Question: Which eye of the girl is closed?, Options: A. Left; B. Right. \n Output: A. Left \n Example2: ... \n Example3: ... \n Input: Image: <image> \n Question: {question}, Options: {options}. \n Output:

Table 8: Task prompts for open-and closed-source MLLMs.